MSD MANUAL
Consumer Version

Ear Ringing or Buzzing

(Tinnitus)

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Ringing in the ears (tinnitus) is noise originating in the ear rather than in the environment. It is a symptom and not a specific disease. Tinnitus is very common—10 to 15% of people experience it to some degree.

The noise heard by people with tinnitus may be a buzzing, ringing, roaring, whistling, or hissing sound and is often associated with hearing loss. Some people hear more complex sounds that may be different at different times. These sounds are more noticeable in a quiet environment and when people are not concentrating on something else. Thus, tinnitus tends to be most disturbing to people when they are trying to sleep. However, the experience of tinnitus is highly individual. Some people are very disturbed by their symptoms, whereas others find them quite bearable.

Subjective tinnitus is by far the most common type. It is caused by abnormal activity in the part of the brain responsible for processing sound (auditory cortex) and is often a symptom of ear-related disorders. Doctors do not fully understand how this abnormal activity develops.

Objective tinnitus is much less common. It represents actual noise created by structures near the ear such as noise generated by blood flowing through arteries or veins. It may be synchronous with the person's heartbeat. Other people can sometimes hear the sounds of objective tinnitus if they listen closely with a stethoscope, and extremely rarely, sometimes just with their own ear.

Causes of Ear Ringing or Buzzing

Subjective tinnitus

Many ear-related disorders include tinnitus as a symptom, and people who have hearing loss, regardless of cause, often develop tinnitus. These disorders include

- Exposure to loud noises or explosions (acoustic trauma)
- Aging (presbycusis)
- Certain medications that damage the ear (ototoxic medications)
- Meniere disease
- Migraines

Tinnitus can also be a symptom in people with middle ear infections, disorders that block the ear canal (such as an external ear infection [otitis externa], excessive ear wax, or foreign bodies), problems with the eustachian tube (which connects the middle ear and the back of the nose) due to allergies or other causes of obstruction, otosclerosis (a disorder of excess bone growth in the middle ear), and temporomandibular disorders. An uncommon but serious cause is a vestibular schwannoma, a noncancerous (benign) tumor of part of the nerve leading from the inner ear.

Objective tinnitus

Objective tinnitus usually involves noise from blood vessels near the ear. In such cases, the sound comes with each beat of the pulse (pulsatile). Causes include

- Turbulent flow through the carotid artery or jugular vein
- Certain middle ear tumors that are rich in blood vessels
- Malformed blood vessels of the membrane covering the brain
- Narrowing or blockage of the venous drainage system of the brain
- Increased pressure within the skull (idiopathic intracranial hypertension)
- Missing bone over the top part of the superior semicircular canal
- Missing bone over one of the veins behind the ear
- Middle ear fluid amplifying sounds of blood flow

The most common noise is the sound of rapid or turbulent blood flow in major vessels of the neck. This abnormal blood flow may occur because of a reduced red blood cell count (anemia) or a blockage of the arteries (atherosclerosis) and may be worsened in people with poorly controlled high blood pressure (hypertension). Some small tumors of the middle ear called glomus tumors are rich in blood vessels. Although the tumors are small, they are very near the sound-receiving structures of the ear, and blood flow through them can sometimes be heard (only in one ear). Sometimes, blood vessel malformations that involve abnormal connections between arteries and veins (arteriovenous malformations) develop in the membrane covering the brain (the dura). If these malformations are near the ear, the person sometimes can hear blood flowing through them.

Less commonly, spasms of muscles of the palate or the small muscles of the middle ear cause clicking sounds. These sounds do not follow the beat of the pulse. Such spasms often have no known cause but may be due to tumors, head injury, or diseases that affect the covering of nerves (for example, multiple sclerosis).

Evaluation of Ear Ringing or Buzzing

Not all tinnitus requires evaluation by a doctor. The following information can help people decide whether a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

Certain symptoms and characteristics are cause for concern. They include

- Tinnitus in only one ear
- Any neurologic symptoms (other than hearing loss), particularly difficulty with balance or walking, but also vertigo or difficulty seeing, speaking, swallowing, and/or talking

When to see a doctor

People with warning signs should see a doctor right away. People without warning signs in whom tinnitus recently developed should call their doctor, as should people with pulsatile tinnitus. Most people with tinnitus and no warning signs have had tinnitus for a long time but should see their doctor about it if they have not already done so.

What the doctor does

In people with tinnitus, doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination may suggest a cause of the tinnitus and the tests that may need to be done (see table Some Causes and Features of Tinnitus).

During the medical history, doctors ask about the following:

- The nature of the tinnitus, including whether it is in one or both ears and whether it is constant or pulsatile
- Whether the person has any neurologic symptoms
- Whether the person has been exposed to loud noise or to medications that can affect the ears

During the physical examination, doctors focus on examining the ears (including hearing) and the neurologic system. They also listen with a stethoscope over and near the person's ear and on the neck for sounds of objective tinnitus. Doctors may do tuning fork tests to evaluate hearing.

Some Causes and Features of Tinnitus

Cause	Common Features*	Diagnostic Approach
Subjective tinnitus (typically a constant tone and sometimes accompanied by some degree of hearing loss)		
Acoustic trauma (noise-induced hearing loss)	History of occupational or recreational exposure to noise Hearing loss	Audiogram
Aging (presbycusis)	Progressive hearing loss, often with family history	Audiogram
Barotrauma (ear damage due to sudden pressure change)	Clear history of exposure to increased air or water pressure	Audiogram
Brain tumors (such as vestibular schwannoma or meningioma) or disorders such as multiple sclerosis or stroke	Tinnitus and often hearing loss in only one ear Sometimes other neurologic abnormalities	Audiogram Gadolinium-enhanced MRI
Eustachian tube dysfunction	Often a long history of decreased hearing and frequent colds, and problems clearing ears with air travel or other pressure change May be in one or both ears (often one ear more of a problem than the other)	Tympanometry Audiogram